MAGNETOM

Maintenance Instructions System

Preventive Maintenance

Aera
Skyra
The protocol M7-010.832.05.01.XX is required for these instructions

Document Version

Siemens reserves the right to change its products and services at any time.

In addition, manuals are subject to change without notice. The hardcopy documents correspond to the version at the time of system delivery and/or printout. Versions to hardcopy documentation are not automatically distributed.

Please contact your local Siemens office to order a current version or refer to our website http://www.healthcare.siemens.com.

Disclaimer

Siemens provides this documentation “as is” without the assumption of any liability under any theory of law.

The content described herein requires superior understanding of our equipment and may only be performed by qualified personnel who are specially trained for such installation and/or service.

Copyright

1.1 General information

To ensure the continuous safety, quality, and reliability of these products, periodic maintenance must be performed in accordance with the Maintenance Instructions / Safety-related tests.

Periodic maintenance includes:

- Safety-related tests (not included in these Maintenance Instructions)
- Preventive maintenance
- Image quality assurance

Image quality assurance should always be performed upon completion of maintenance activities.

1.2 Safety Information

WARNING

Please follow the general safety guidelines as well as the MR-specific safety information!

Failure to comply with the safety notes may result in death or serious injury.

» **Read the safety notes contained in the general safety notes as well as the MR-specific safety information in detail prior to starting work. Adhere to the information provided at all times.**

1.3 Documentation

A protocol of each maintenance activity must be generated. All fields must be completed.

Each work step or test performed successfully has to be marked “OK” in the test protocol. The assignment n.a. (not applicable) indicates that the checkpoint or measured value is not used for this system. If an error occurs that cannot be eliminated, mark the respective test with “not OK”.

The Maintenance Protocol has to be archived by the operator in the System Owner Manual. We recommend that country organizations also compile such documentation to ensure optimum product operation, e.g. should damage occur.

1.4 Acronyms:

Abbr.	Explanation
SI	Safety Inspection
SIE	Safety Inspection Electrical Safety
SIM	Safety Inspection Mechanical Safety
PM	Preventive Maintenance
PMP	Preventive Maintenance Preventive Parts Exchange, External Inspection, etc.
PMA	Preventive Maintenance Adjustments
PMF	Preventive Maintenance Function, Operating Value Check
Q	Quality Check
QIQ	Quality Check Image
QSQ	Quality Check System
SW	Software Maintenance

1.5 Time Frame for the Maintenance Schedule

Tab. 1 Maintenance Schedules

Component / activity	Initial[1] (months)	Interval[2] (months)	Time required (minutes)	Required tools and parts
Computer	18	12	10	Vaccum cleaner
TFT monitor	18	12	10	
Vacuum pump	18	24	5	Filter p.n. 10496420
Gradient filters	18	12	10	
Cooling	18	12	15	
SEP (optional)	18	12	20	50/41 mm open-end wrench
Cold head	18	12	5	
Adsorber	36	36	70	Adsorber p.n.7461127
Patient table	18	12	20	If required hydraulic oil material number 10683734
Tim Dockable Table	18	12	15	
RF room door	18	12	10	10 cm wide strip of paper
Software	18	12	30 - 60 [3]	CD-R with medical grade
QA	18	12	90 [4]	The QA is requested twice a year (once QA during maintenance and once QA during safety -related tests)
Annual average time including all options[5] : 261 min (4.4)				

1. Initial maintenance means the first maintenance schedule after installation of the system
2. Interval means the recurrent maintenance interval after the initial maintenance
3. Depending on data quantity (calculation with 30 min)
4. The required time for QA (70 - 110 min.) depends on the installed options. Calculation within 90 min. during maintenance
5. Without Combi Dockable Table, without Cold head exchange

Tab. 2 Combi Dockable Table Maintenance Schedules

Component / activity	Initial[1] (months)	Interval[2] (months)	Time required (minutes)	Required tools and parts
Combi Dockable Table	18	12	35	Hydraulic oil material number 10683734
Annual average time 35 min (0.6 h)				

1. Initial maintenance means the first maintenance schedule after installation of the system
2. Interval means the recurrent maintenance interval after the initial maintenance

1.5.1 Cold Head

Tab. 3 Cold Head Replacement

Activity	Interval	Time required	Note
Cold head replacement	see footnote[1]	Aera 400 min. Skyra 540 min.	Service personnel training on the Sumitomo cold head is offered by SMT. Instructions for cold head replacement are shown in the video “Sumitomo Refrigerator System”. The video is a supplement to the SMT course.

1. The cold head has to be replaced latest every 24 months unless a daily remote monitoring of the magnet refrigerator status is performed in which case a condition based exchange at a later stage is possible..

2.1 System Status

- Switch System off

In case to access to the components installed on the magnet the appropriate service covers have to be removed, please refer to REP - System/Covers.

2.2 Computer MRC (MRAWP) - MRSC (MRWP)

Required time: 10 min

Required tools: Vacuum cleaner

PM Cleaning the computer

MRC

- Check all air inlets and outlets for dust. If necessary, clean the computer with a vacuum cleaner.

MRSC if installed

- Check all air inlets and outlets for dust. If necessary, clean the computer with a vacuum cleaner.

Fig. 1: Computer

Fig. 2: Computer

2.3 TFT monitor

Required time: 10 min

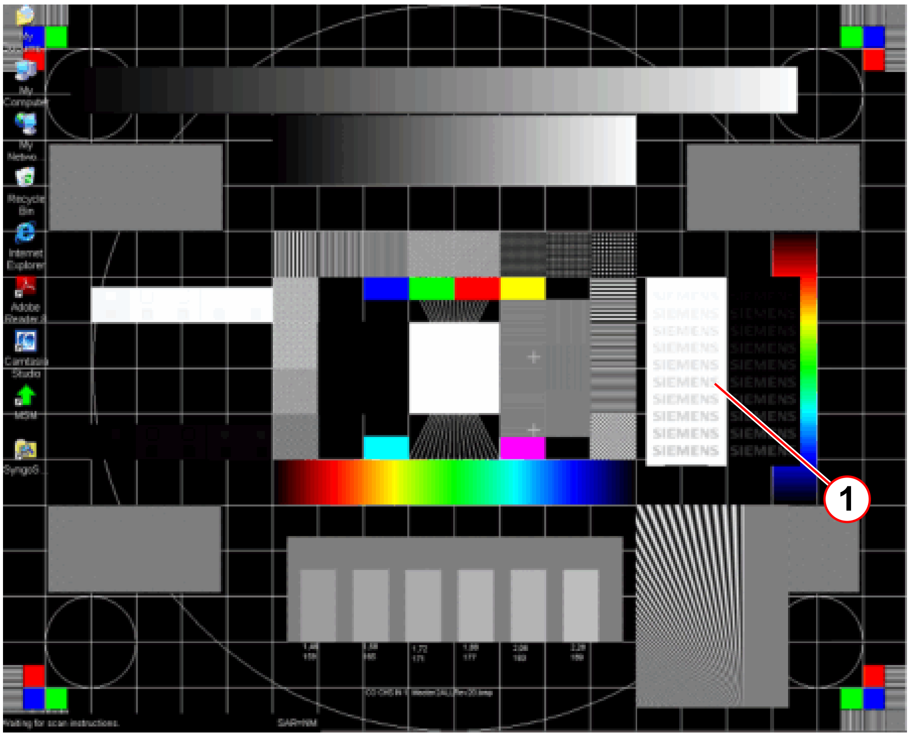
Required tools: n.a.

Prerequisites for VE11C: The **Advanced User** has been enabled (**Service Software** > **Configuration** > **Security** > **Advanced User** > **Enable Advanced User**) and the Advanced User password must be available.

2.3.1 Calling up the test image

- Simultaneously press buttons **Ctrl** and **Esc**.
  - » the Windows **Start Menu** appears
- Click: All Programs > Advanced User > enter password > press OK button
- Simultaneously press **Ctrl** + **Shift** + **Esc**
- Select the Task Manager and choose the **Applications** task card.
- Right-click on the CapGM-xxxx in the list and select **Minimize**.
- Minimize the Task Manager (WIN Desktop is now displayed in the foreground)
- Right-click on the desktop background
- Click **Personalize** and choose **Desktop Background**
- Click Browse
- Select path: Computer > C:\MedCom\utils\Imagequality
- Select file Master2AllRev20.bmp
- Click **save changes**.
- If the syngo screen appears again, select the task manager (Alt + Tab) then right click on the CapGM-xxxx and select Minimize.

Fig. 3: Test image - master2AllRev.20



(1) Siemens test logo

2.3.2 Check

PMA Checking the TFT monitor

- Check the MRC TFT monitor using the test image.
  - » All colors are displayed
  - » Gray scale is correctly displayed
  - » Geometry is correct
  - » 10 SIEMENS test logos are visible in the black and white field
- If the MRSC TFT monitor (option) installed, check the MRSC TFT monitor following the same procedure.
- If the In-Room MRC TFT monitor (option) installed, check the In-Room MRC TFT monitor first following the same procedure.

If any adjustment is necessary, refer to 'Replacement of Parts REP'; 'Visually inspecting/ evaluating the TFT monitor' in detail.

2.3.3 Set the normal mode

- Right-click on the desktop background
- Click **Personalize** and choose **Desktop Background**
- Select **Solid Colors** in the **Picture Location** menu.

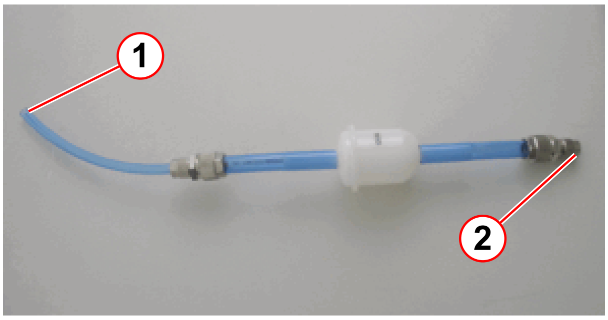
- Select color black
- Click **save changes**.
- Close the **Personalization** window and the **Task Manager**
- Simultaneously press **Alt** + **Tab** until syngo screen appears.

2.4 Vacuum pump (comfort kit option)

Required time: 5 min

Required tools: Filter (material number: 10496420)

Fig. 4: Vacuum pump filter

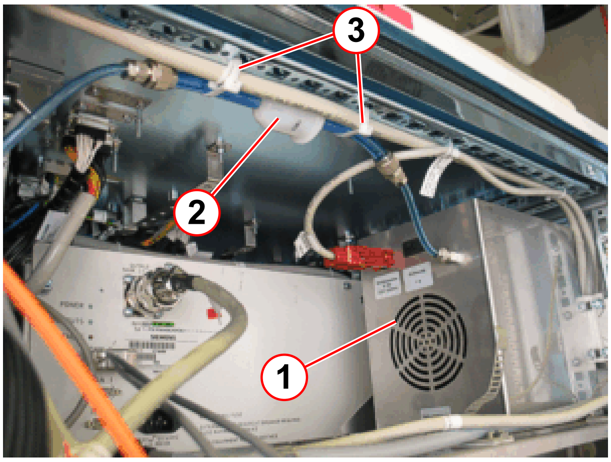


(1) Connection - vacuum pump
(2) Connection - hose to the examination room

PMP Replacing the vacuum pump filter (every 2 years)

The vacuum pump is an option (comfort kit) mounted in the upper section of the EPC.

Fig. 5: Upper part of the EPC

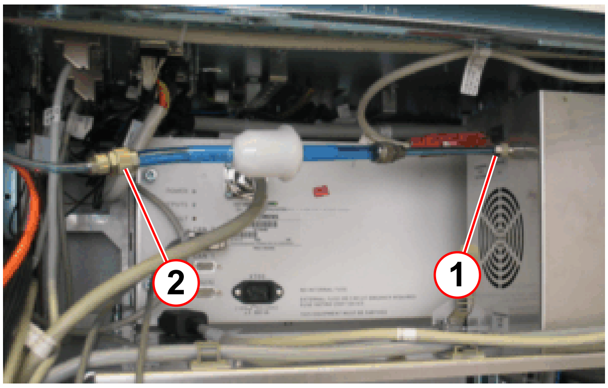


(1) Vacuum pump
(2) Vacuum pump filter
(3) Cable ties

The vacuum pump filter is fixed on the metal frame of the EPC with cable ties.

- Cut the cable ties.

Fig. 6: Upper section of the EPC



(1) Coupling nut
(2) Coupling nut

- Loosen the coupling nut (1) at the housing of the filter pump.
- Remove the hose.
- Remove the coupling nut and place it on the hose of the new filter.
- Connect the new filter to the vacuum pump and mount it with the coupling nut.
- Loosen the coupling nut (2) of the hose to the examination room.
- Remove the old filter.
- Connect the new filter to the hose of the examination room and mount it with the coupling nut.

- Fix the vacuum pump on the metal frame with cable ties.
- Enter the replacement date in the Maintenance Protocol.

2.5 Gradient filters

Required time: 10 min

Required tools: n.a.

The gradient filters are ventilated by 3 fans per axis.

PM Checking the gradient filter fans

Fig. 7: Gradient filter fans

» If one or 2 fans per axis are defective, replace the fan assembly.

» If all 3 fans per axis are defective, the corresponding gradient filter has been damaged due to overheating. Replace the fan assembly **and the corresponding gradient filter.**

2.6 Cooling

Required time: 15 min

Required tools: n.a.

PMP Cooling system - general checks

Check the entire water system for leaks and condensation, especially all water connections and water hoses within/at the:

- EPC
- GPA
- ACS
- SEP or IFP (option)
- Magnet

- Open > **Service Software** > **Magnet & Cooling** > **Cooling Status**
- Check the cooling status of the system
  - EPC
  - SEP or IFP - Chiller
  - RFCEL
  - Coil Temperature Sensors
  - TX Box
  - Equipment room
  - Magnet room

2.7 SEP (option)

Required time: 20 min

Required tools: 50mm or 41 open-end wrench, needle nose pliers, cloth to absorb water, bucket, flash light

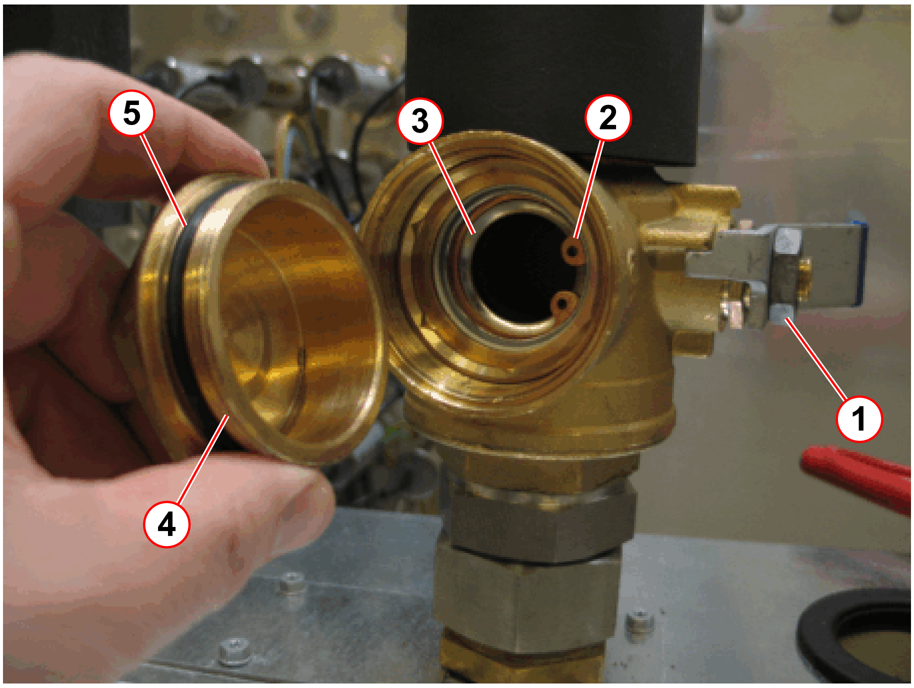
> Cooling water containing ethylene glycol must be collected while draining. Return the cooling water to the water circuit during the next filling.
>
> Avoid contact with eyes and skin.

PMP Checking the water circuit of the SEP

Cleaning the strainer of the **primary water circuit** (once a year).

- Switch off the system.
- Check for leaks.
- Slowly close the valve (→ 1/Fig. 8 Page 19).

Fig. 8: SEP - valve with strainer



(1) Handle - closed
(2) Circlip
(3) Strainer
(4) Screw cap
(5) O-ring

> The closed valve contains approximately 200 mL cooling water. Use a bucket to collect the water.

- Remove the screw cap (➔ 4/Fig. 8 Page 19) slowly by using a open-end wrench.
- Use a bucket to collect the water.
- Use a cloth to absorb some water from the strainer hole.
- Remove the circlip (➔ 2/Fig. 8 Page 19) by using needle nose pliers.
- Remove the strainer (➔ 3/Fig. 8 Page 19).
- Inspect the strainer for contamination and clean it by rinsing under tap water.

Fig. 9: Strainer

Reassemble the valve in reverse order.

- Install the strainer.
- Mount the strainer with the circlip.
- Check the O-ring (➔ 5/Fig. 8 Page 19) of the screw cap.
- Screw the screw cap into the shut-off valve.
- Slowly open the shut-off valve.
- Check for leaks on the screw cap.

Final steps

- Switch on the system.
- Open > **Service Software** GUI > **Magnet & Cooling** > **Cooling Status**
- Check the **Primary Water - Temperature and Flow**.
- If necessary, advise the customer to initiate corrective action immediately.

Secondary water circuit

- Open > **Service Software GUI** > **Magnet & Cooling** > **Cooling Status**

- Check the water flow and pressure of the secondary water circuit.
  - » All data is within the specifications.
  - » If **SEP - Return Pressure** < **0.9 bar** (recommended pressure value for refilling), **refill water**.
  - » If the error message `Inverter overrun` appears, clean the strainer.
  - » If the water flow and/or the pump revolution is outside of specifications, clean the strainer of the secondary water circuit.

2.8 Magnet system

2.8.1 Cold Head

Required time: 5 min

Required tools: n.a.

The cold head has to be replaced latest every 24 months unless a daily remote monitoring of the magnet refrigerator status is performed in which case a condition based exchange at a later stage is possible.

PM Cold head is monitored (cold head replacement on demand)

PM Replacing the cold head (w/o cold head monitoring replacement every 24 months)

The cold head has to be replaced every 24 months according to OEM recommendation.

> Special training is required for the cold head replacement. The cold head replacement procedure is not included in this description. During the cold head replacement the magnet pressure switch must be checked.

- After the cold head is replaced, enter the replacement date in the Protocol.
- Confirm the execution of the magnet pressure switch check in the Protocol.

2.8.2 Adsorber

PM Replacing the adsorber (F70; every three years)

2.8.2.1 Replacing the adsorber (F70; every three years)

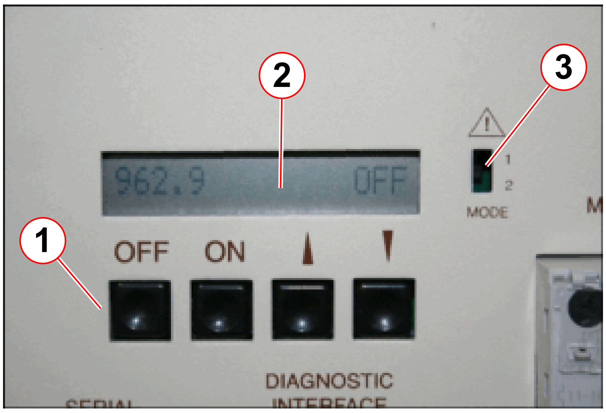
Required time: 70 min

Required parts and tools:

- Adsorber part number (**07461127;** includes venting adaptor)
- #2 Phillips screwdriver
- F70 Spanner kit **10125650** (delivered with the system)

> Adsorbers are filled with 99.999 grade helium to a pressure of 14-16 bar. Do not vent the new adsorber.

Fig. 10: F70 System status display



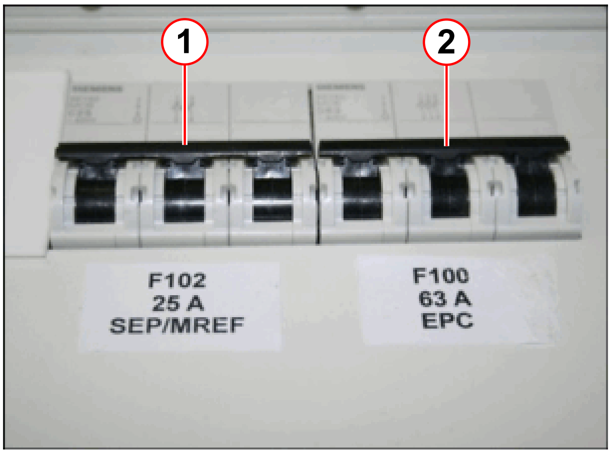
(1) Control buttons
(2) LCD display
(3) Configuration selector switch

- If the **configuration selector switch** has been set to **configuration 2**, both **ON** and **OFF** buttons are **disabled.** (the compressor is controlled via the pressure within the magnet).
  - Switch the compressor **ON** or **OFF** using the **main power switch**.
- If the compressor is set to **configuration 1**, use the **ON/OFF** buttons on the control panel.

Adsorber removal

- The compressor may be a **stand-alone** unit, or fitted into an **SEP cabinet**.
- **Press** the **OFF** button on the **control panel**.
- Turn **off** the compressor at the **main power switch.**

Fig. 11: EPC Cabinet - circuit breakers

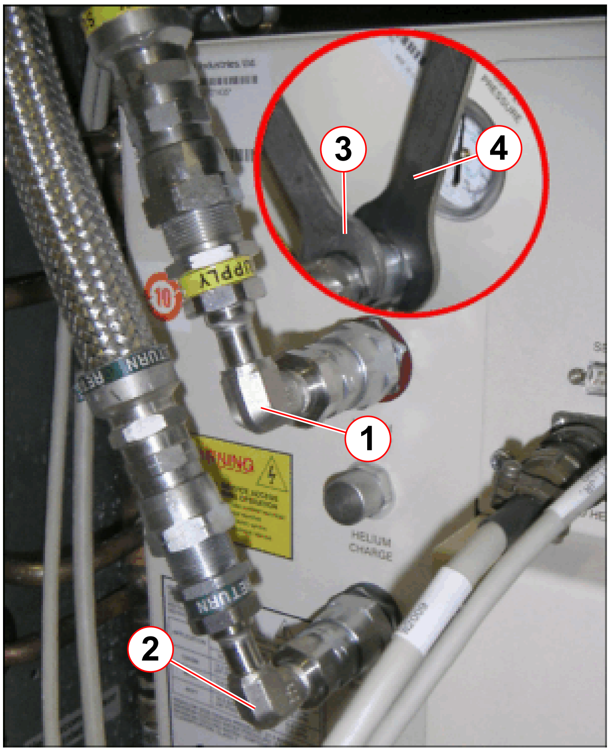


(1) F102 - Isolates SEP and refrigerator
(2) F100 - Isolates EPC cabinet

- **Isolate power** to the compressor:
  » **EPC cabinet** - SEP or IFP (stand alone unit) - switch off circuit breaker **F102**.

- The **alarm box** will display a warning LED and an **audible alarm** will sound.
- **Press** the **AUDIO ALARM OFF** button to cancel the audible alarm.

Fig. 12: F70 He compressor connections



(1) SUPPLY (high pressure) 90-degree elbow
(2) RETURN (low pressure) 90-degree elbow
(3) Hold nut (1" spanner/wrench)
(4) Turn nut (1 3/16" open end crowfoot spanner/wrench)

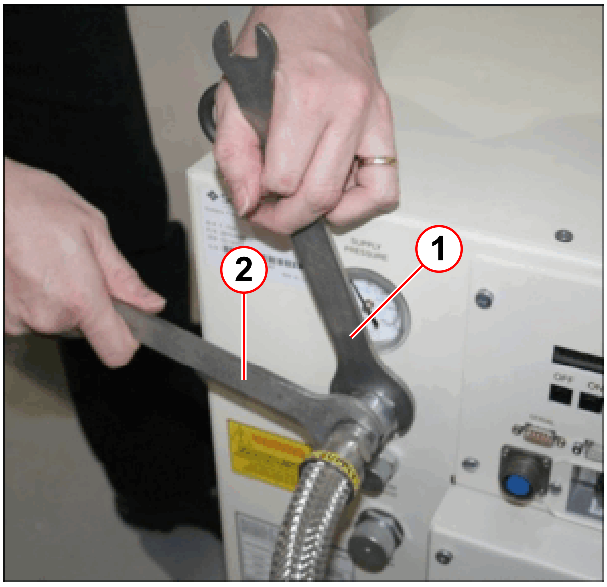
SEP cabinet option

Each **helium pressure line** is connected to the compressor via a **90 degree elbow**.

- **Remove** the self-sealing helium pressure lines in the following order:
  - **Return** (low pressure)
  - **Supply** (high pressure)
- **Hold** nut (3) and **turn** nut (4).

> ℹ 90° elbow connections are not required for a stand-alone unit.

Fig. 13: F70 He compressor connections (stand-alone unit)

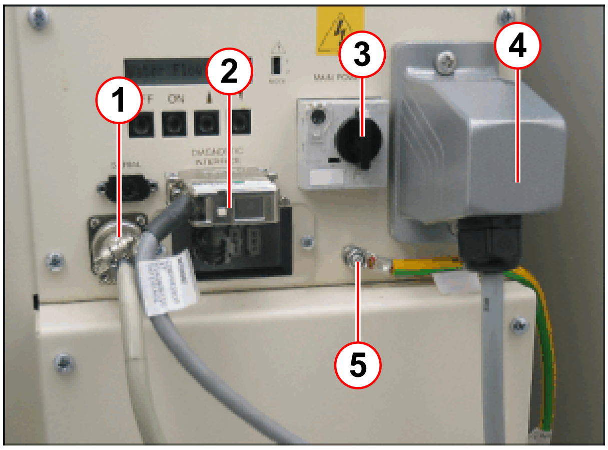


(1) Turn nut (1 3/16" open-end crowfoot spanner/wrench)
(2) Hold nut (1" spanner/wrench)

Stand-alone unit

- **Remove** the self-sealing helium pressure lines in the following order:
  - **Return** (low pressure).
  - **Supply** (high pressure).
- **Hold** nut (2) and **turn** nut (1).

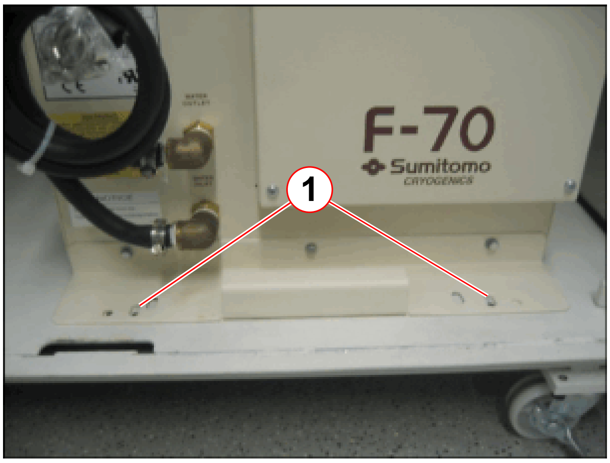
Fig. 14: F70 Electrical cable connections



(1) Cold head cable
(2) MSUP DB25 Diagnostic interface cable
(3) Main power switch
(4) Mains supply cable
(5) Ground wire

- **Disconnect** the **line power supply, cold head motor,** and **MSUP** interface cables.
- **Disconnect** the **ground wire**.

Fig. 15: F70 Compressor fixing plate



(1) M8 Hex hd screws (2 off)

SEP cabinet option:

- **Remove** the 2 M8 hex head screws securing the **fixing plate** to the cabinet.
- Use the **fixing plate handle** to carefully pull the compressor onto the wooden support 'ramp'.

  The unit is fitted with **nylon slide rails** to simplify movement.

WARNING

Risk of personal injury and damage to equipment.

The weight of the compressor is 100 kg (220 lbs).

» **When installing or removing from the cabinet, reduce the height difference by ensuring there is a sturdy support 'ramp' in position to support the unit.**

NOTICE

Prevent equipment failure.

» **To prevent oil from flowing into unwanted places, do not tip the compressor more than 10 degrees from the horizontal.**

Fig. 16: Supporting ramp

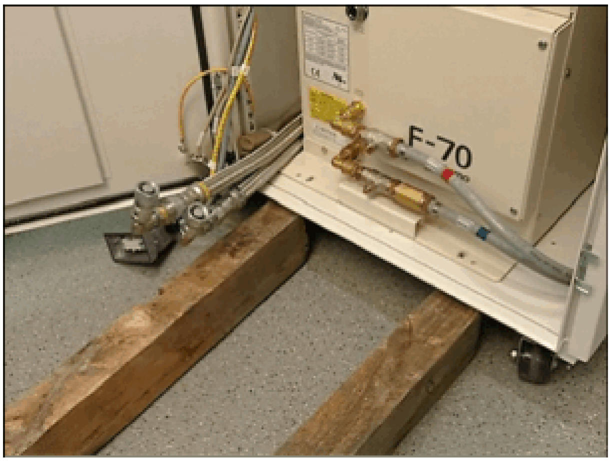


Fig. 17: Move the compressor onto the supporting ramp

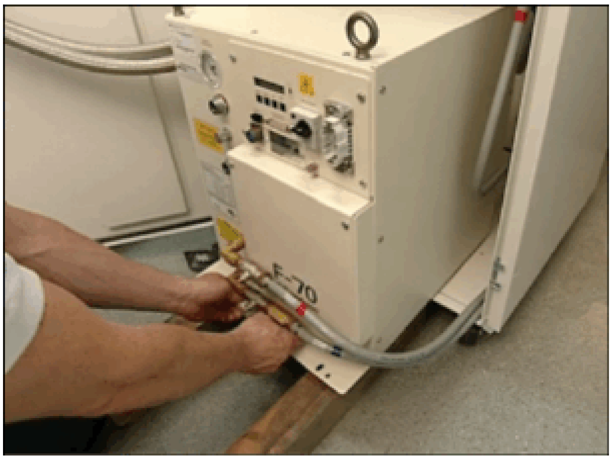
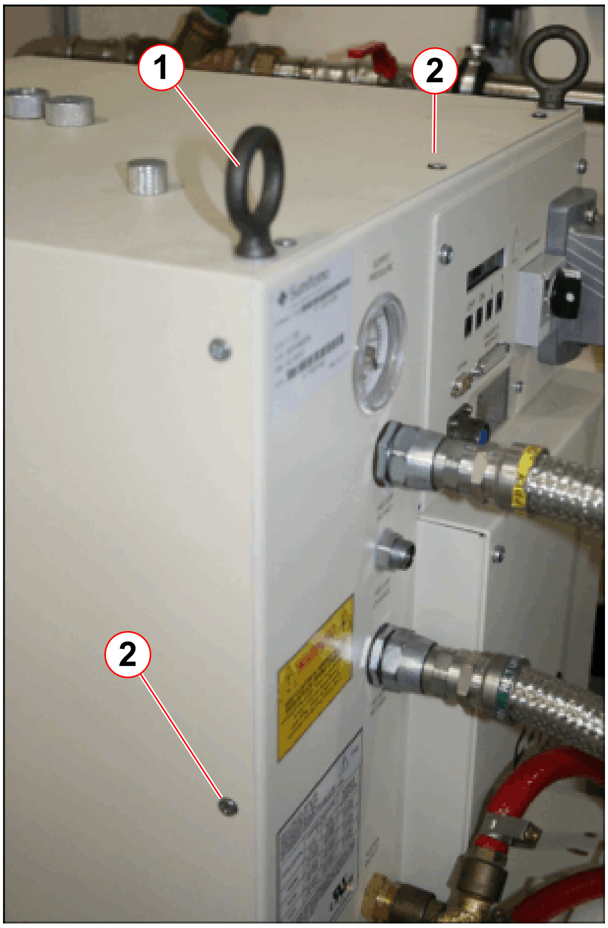


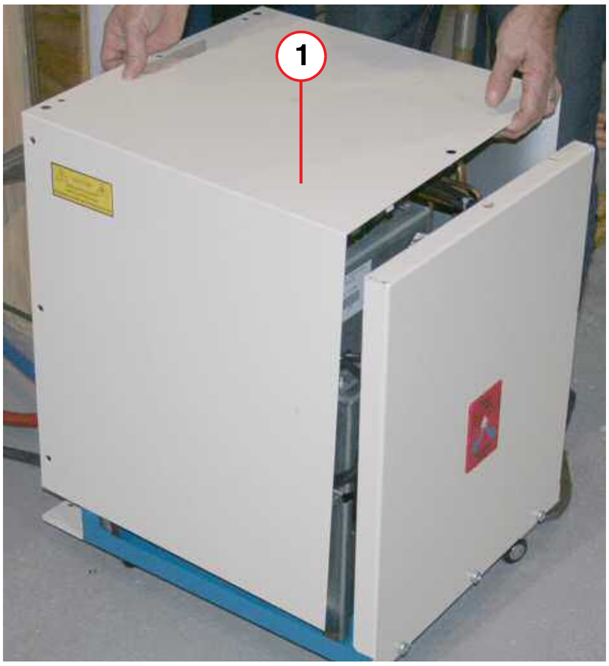
Fig. 18: F70 Lifting eyebolts and cover screws



(1) Lifting eyebolts (3 positions)
(2) Cover panel securing screws

- **Remove** the **3** lifting **eyebolts** and all cover panel **screws**.

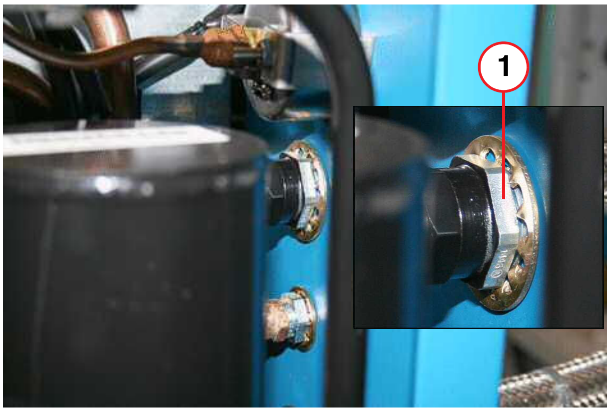
Fig. 19: F70 Removing/replacing the cover



(1) U-shaped cover

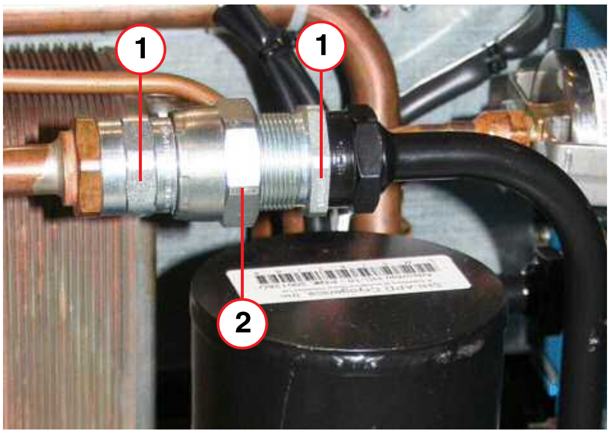
- **Lift off** the U-shaped **cover panel.**

Fig. 20: F70 Adsorber - high pressure supply coupling (inlet side)



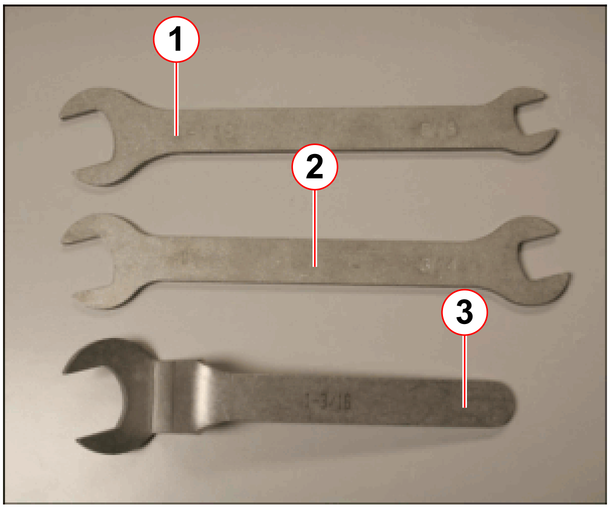
(1) HOLD stationary nut (1” open-end spanner/wrench)

Fig. 21: F70 Adsorber - oil separator Aeroquip coupling



(1) HOLD stationary nut (1” open-endspanner/ wrench)
(2) TURN moveable coupling (1 3/16” open-end crow foot)

Fig. 22: F70 Spanner kit (part no. 10125650)

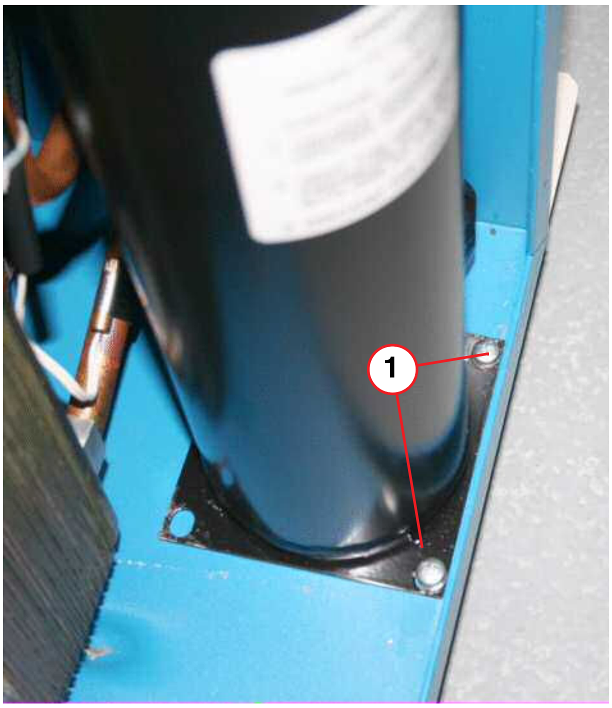


(1) 1 1/8" open-end and 5/8" open-end spanner/wrench
(2) 1" open-end and 3/4" open-end spanner/wrench
(3) 1 3/16" open-end crowfoot spanner/wrench

- Always use the three open-ended wrenches when connecting and disconnecting the couplings.
  - **Hold** the stationary nut(s).
  - **Turn** the moveable coupling.

- **Disconnect** the self-sealing coupling on the **inlet side** of the adsorber: (→ Fig. 20 Page 28).
  - **Hold** the stationary nut (1) on the **inlet side**.
  - **Turn** the **high pressure locknut** on the front panel (→ 1/Fig. 24 Page 30).
- **Disconnect** the adsorber at the **oil separator coupling** (→ Fig. 21 Page 28).

Fig. 23: F70 Adsorber base securing screws



(1) Securing screws (2 off)

- **Remove** the 2 **base securing screws**.

Fig. 24: F70 High pressure Aeroquip coupling

(1) Locknut
(2) Flat rubber gasket
(3) Nylon washer

- **Remove** the high pressure **locknut** and **nylon washer**.

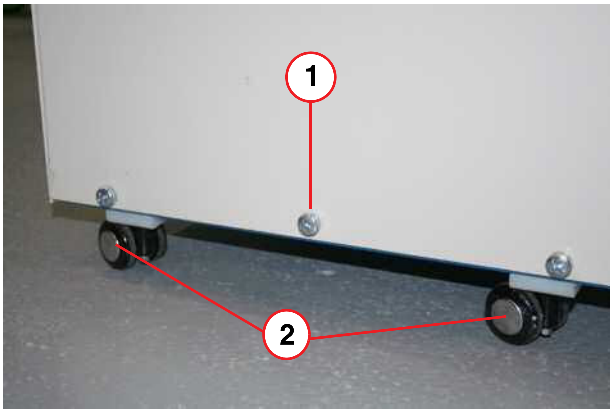
- Carefully **remove** the adsorber.
- **Retain** all removed **attachments**.

> Adsorbers are filled with 99.999 grade helium to a pressure of 14-16 bar. Do not vent the new adsorber.

Adsorber replacement

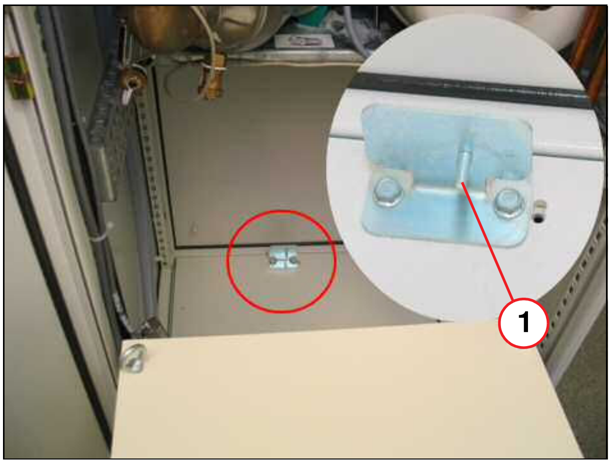
- **Remove** the **caps** from the gas lines of the new adsorber.
- **Install** the **lock washer** on the **supply coupling** of the new adsorber.
- **Insert** the **supply coupling** through the **front panel** and position the adsorber.
- **Secure** the adsorber to the **base** using the **screws** (→ 1/Fig. 23 Page 29).
- **Install** the **nylon washer** and **locknut** on the high pressure **supply** coupling (→ Fig. 24 Page 30).
- **Connect** the adsorber at the **oil separator coupling** and perform a leak check. (→ Fig. 21 Page 28).
- **Reinstall** the **cover panel** using the retained **screws**.

Fig. 25: F70 Back panel view



(1) Center screw is removed for SEP cabinet installation
(2) Casters x 4 (stand-alone units only)

Fig. 26: SEP - Compressor guide pin

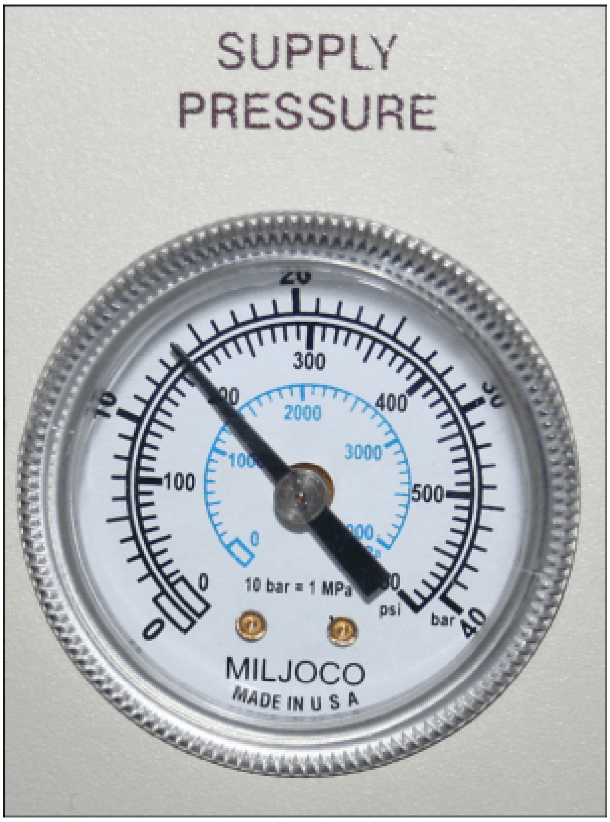


- Slide the compressor into the **SEP cabinet** and locate the **guide pin** in the **compressor hole.**
- **Resecure** the **fixing plate** using the 2 M8 hex head screws.

> Tighten all Aeroquip screw connections to the end stop and turn them back by 1/4 turn.

- **Reconnect** the high pressure (**supply**) and low pressure (**return**) helium gas lines (→ Fig. 12 Page 24).
- Reconnect the **cold head cable, diagnostic cable, and power cable** and **ground wire** (→ Fig. 14 Page 25).

Fig. 27: F70 Static pressure reading



- **Check** the **equalization (static)** pressure on the pressure gauge.
  » **Gas charging** will be required if pressure is below expected.

Equalization (static) pressure at 20 ^{0}C (680 F) for gas lines 12 to 20 meters long

Tab. 4 Helium gas pressures

Cold head	Static pressure	Dynamic pressure
4 K - 50/60 Hz	13.5 - 14.0 bar (13.6 nominal) 196 - 203 psi (G) 1350 - 1400 kPa	21 - 22 bar 304 - 319 psi (G) 2100 - 2200 kPa

- Switch **on F102** (EPC cabinet).
- Switch **on** the compressor at the **main power switch**.
- **Configuration 1 only** - switch **on** the compressor using the **ON** button on the **control panel**.
- Check the **alarm box status** and **clear** any **faults**.
- Check the gas Aeroquip **couplings** for **leaks**.

CAUTION

The adsorber is charged with helium gas!

Improper handling of the used adsorber may cause injury.

» **Follow the venting procedure for safe depressurization and disposal of the used adsorber.**

Safe disposal of the used adsorber

A **venting adapter** is included with the new adsorber.

- **Attach** the **adapter** to one of the **self-sealing couplings** on the used adsorber.
- **Vent** the used adsorber to **atmospheric pressure.**
- Give the **used adsorber** to a **disposal company.**

2.9 Patient table

2.9.1 Hydraulic system of the patient table

Required time: 20 min

Required tools and parts: 2.5 mm Allen key, 3 mm Allen key,

if required hydraulic oil material number 10683734

PM Checking the hydraulic system of the patient table

- Move the table into the home position.
- Lower the table prior to loosening the bolts that hold the upper flanges of the cover for the lifting column to reduce stress on the flange.
- Loosen the cover of the lifting column on the upper fastening and lower the cover.
- Fixed table; remove the foot cover
- Dockable table; remove the cover of the chassis on the front and foot end
- Remove the foot cover; magnet side (Dockable Table only).
- Check the hydraulic system for damage and leaks.
  - Hydraulic cylinder for lifting scissors
  - Hydraulic piston for lifting scissors
  - Oil reservoir
  - Hydraulic lines and connections
  - Hydraulic cylinder for the docking mechanism (Dockable Table only).
  - Hydraulic piston for the docking mechanism (Dockable Table only).

  » The hydraulic system has no damage or leaks.
- Move the table all the way down.

  » The table is in the undermost position.
- Check the hydraulic oil level on the oil reservoir.

  » The hydraulic oil level is within the minimal - maximal marker.

  » Only a leakage causes oil loss. Troubleshooting is necessary.
- Mount the covers.

2.9.2 Option dockable table: docking station

Required time: 15 min

Required tools: n.a.

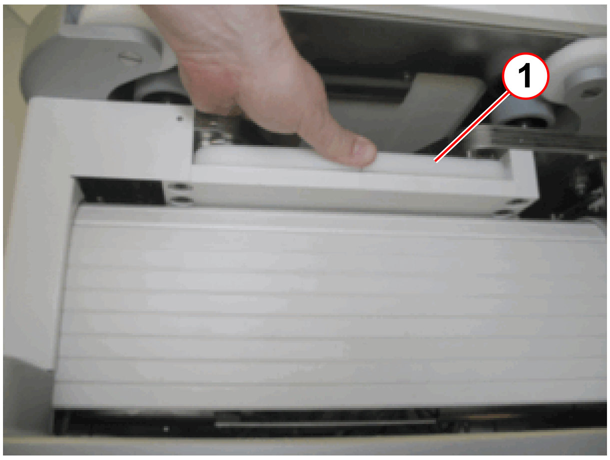
PM Checking the docking station (dockable table only)

- Move the table approximately 300 mm into the magnet.
- Step on the docking / undocking and the up / down pedal.

  » Stepping on the pedals has no effect on the table.

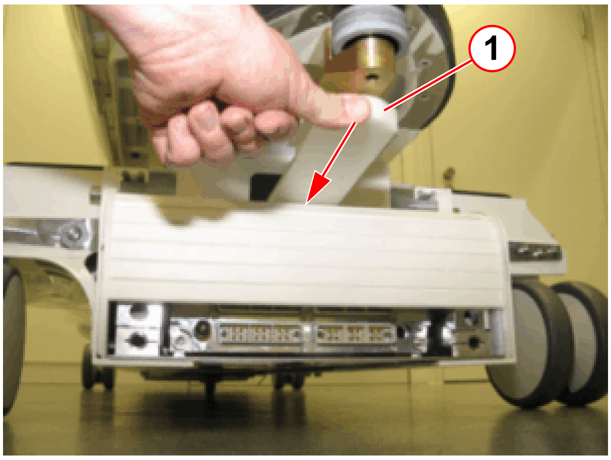
- Move the table all the way out (home position).
- Step on the docking / undocking and the up / down pedal.
  » The table is undocked.
- Slide back the roller blind (➔ Fig. 28 Page 35)(➔ Fig. 29 Page 35).
- Check the roller blind of the docking station and of the dockable table for wear and tear.
- Check the function and mobility of the roller blinds.

Fig. 28: Docking station - roller blind



(1) Lever to move the roller blind

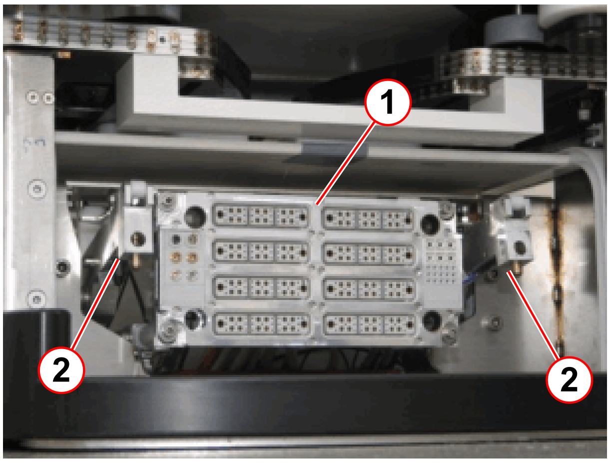
Fig. 29: Patient table - roller blind



(1) Lever to move the roller blind

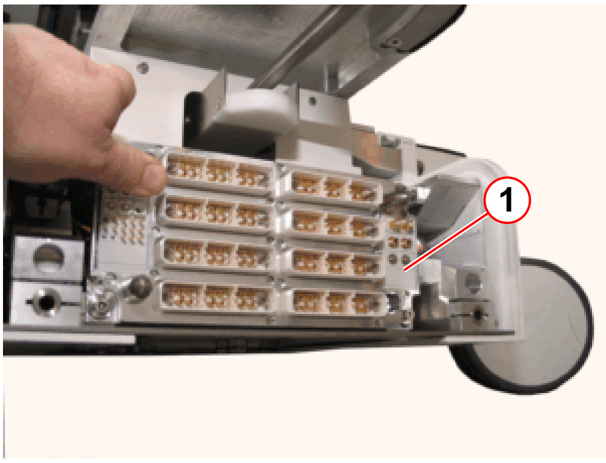
The connector assemblies are movable mounted

Fig. 30: Patient table docking port on the magnet



(1) Connector assembly
(2) Locking hook

Fig. 31: Patient table docking port



(1) Connector assembly

- Check the mobility of the connector assemblies (➔ Fig. 30 Page 35).
  » The connector assemblies of the patient table and the docking station are free to move.

- Check the connectors for wear and tear.
    - » The connectors must not be damaged
- Clean the connectors by using the brush and the non-fuzzing cloth.
- Check the locking hooks (→ 2/Fig. 30 Page 35) for wear and tear.
    - » The locking hooks must not be damaged.

2.10 RF room door

Required time: 10 min

Required tools: 10 cm wide strip of paper

2.10.1 Door

PMP Checking the RF room door

1. Check contact springs for mechanical damage. The contact springs should bounce back. Test the line contact at the outer surface of the contact by inserting a 10 cm wide strip of paper and pulling it out. Repair minor damage using appropriate tools, such as flat nose pliers.
2. The metal of the contact surfaces and contact springs has to be clean. Remove any dirt, contamination, or hardened grease.
3. Check the hinges for cracks or possible damage.

In case of a defect (e.g. broken or missing contact springs, damaged hinges), inform the customer to initiate corrective action immediately.

2.11 Quality Assurance (QA)

Required time: 70 - 110 min depending on installed options.

Required tools: QA Phantom Setup

The quality measurement of the body coil is a fully automated procedure. The results are entered in the on-site database.

> Use the Online Help to obtain detailed information regarding, e.g., table control, phantom positioning, or specific items and procedures. Press the 'Help' button in the header bar of the corresponding platform.

Preliminary steps

Starting the service platform

All actions of the quality assurance procedure are performed via the service platform.

- Start the Service platform **> Options > Service > Local Service**.

PM Quality assurance measurement

Go to the Home menu and press the corresponding button to access the **Quality Assurance** sub platforms.

- Start the measurement by clicking the "Start" button.
  - » The **Quality Assurance** check is successfully completed when every individual QA step indicates the status **< done >**.

> If the results are out of tolerance, perform "Tune-Up" or troubleshooting, if necessary. Repeat "Quality Assurance".

2.12 Software

2.12.1 Cleaning up directories

Required time: 5 min

Required tools / parts: n.a.

> Prerequisites for VE11C: The **Advanced User** has been enabled (**Service Software** > **Configuration** > **Security** > **Advanced User** > **Enable Advanced User**) and the Advanced User password must be available.

SW Deleting the MR save log files

Starting the Explorer

- Press <**Ctrl-Esc**>
- Select <**Start**>, < **All Programs**>, <**Advanced User**>, enter the Advanced User password.
- Press <**Ctrl-Esc**>
- Select < **All Programs**>, <**Accessories**>, <**Windows Explorer**>

Delete files

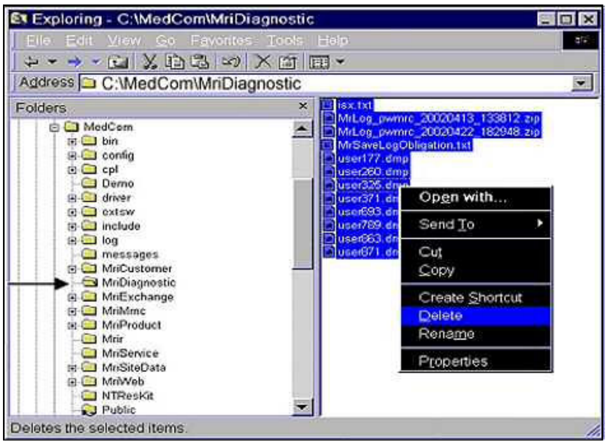


Fig. 32: Exploring C:\MedCom\MriDiagnostic

- Navigate to" **C:\MedCom\MriDiagnostic ⏎**"
- Click once in the right window and select all files <**Ctrl-a**>
- Right-click one of the files and select <**Delete**>.
- This opens the pop-up window "Confirm Multiple File Delete"
- Click <**Yes**> to delete the files
- Close the Windows Explorer

2.12.2 Site-specific data

» Although site-specific data should be saved once a year, we recommend to perform it every three months.

SW Saving site-specific data

Backup of the site-specific data is possible on CD (→ Backup site-specific data on CD / Page 40) and on the USB device (→ Backing up site-specific Data on external USB device / Page 40).

2.12.2.1 Backup site-specific data on CD

Required time: 10-60 min

Required tools / parts: CD-R or USB device

> Use only CD-R 's with medical grade, e.g. Siemens Medical CD-R 80 or TDK Medical CD-R.

Save the site-specific data each time you make changes to the site-specific configuration, tune-up, or sequences and protocols. In case of a hard disk crash, you will be able to restore a valid set of site-specific data.

- Login as **administrator**
- Start the service platform, select **Backup & Restore**.
- Insert a recordable CD-disk in the CD/DVD-burner.
- Select **Backup** in the “Command” pull-down menu.
- In the menu Drives, select the target drive CD/DVD-burner (drive letter **S:**) used to save the data.
- Select **Master-Backup** in the “Packages” menu.
- Click **Go** to start the “Backup”, check for errors in the log-window after the backup has been completed.

Setting that are not part of the site-specific data:
Remote access mode (No Access, Limited Access, etc. in the customer UI).
Auto-transfer flag (“Configuration”, “Service”, “Auto Transfers”)
Postscript Printer Configuration (Customer paper printer)
ImageText Settings (ImageText Settings from ImageText Dialog)
Time zone
Transfer Rules

> During backup at the MRSC, a “**CorRea.bru**” error may be displayed, please ignore this error.

2.12.2.2 Backing up site-specific Data on external USB device

Required time: 10-60 min

Required tools/parts: USB device

> Please check the external medium (e.g. USB-disk) first with the virus scanner of the service PC. Be sure that the external medium used is free of viruses.

- Log in as **administrator** ↵.

- Start the "Internet Explorer".
- Connect the USB device.
- Check what drive letter was set for the USB device with the Windows File Explorer.
- Log in to the service platform and select **Backup & Restore**.
- Select Backup in the Command menu, select the drive letter of the USB device, e.g.,**F** in the Drives menu.
- Select **Master-Backup** in the Package menu, start the backup by clicking **OK**.
- (Estimated time ~10-60 min) Check for errors using the log window.
- Click **Home** to leave Backup & Restore.

2.13 Option: Combi Dockable Table

2.13.1 Option Combi Dockable Table: Hydraulic System

Required time: 20 min

Required tools and parts: 2.5 mm Allen key, 3 mm Allen key,

if required hydraulic oil material number 10683734

PM Option Combi Dockable Table: Checking the hydraulic system

- Move the table into the home position.
- Lower the table prior to loosening the bolts that hold the upper flanges of the cover for the lifting column to reduce stress on the flange.
- Loosen the cover of the lifting column on the upper fastening and lower the cover.
- Remove the cover of the chassis from the head and foot ends
- Remove the foot cover; magnet side
- Check the hydraulic system for damage and leaks.
  - Hydraulic cylinder for lifting scissors
  - Hydraulic piston for lifting scissors
  - Oil reservoir
  - Hydraulic lines and connections
  - Hydraulic cylinder for the docking mechanism.
  - Hydraulic piston for the docking mechanism.

  » The hydraulic system has no damage or leaks.
- Move the table all the way down.

  » The table is in the undermost position.
- Check the hydraulic oil level on the oil reservoir.

  » The hydraulic oil level is within the minimal - maximal marker.

  » Only a leakage causes oil loss. Troubleshooting is necessary.
- Mount the covers.

2.13.2 Option Combi Dockable Table: Docking Station

Required time: 15 min

Required tools: n.a.

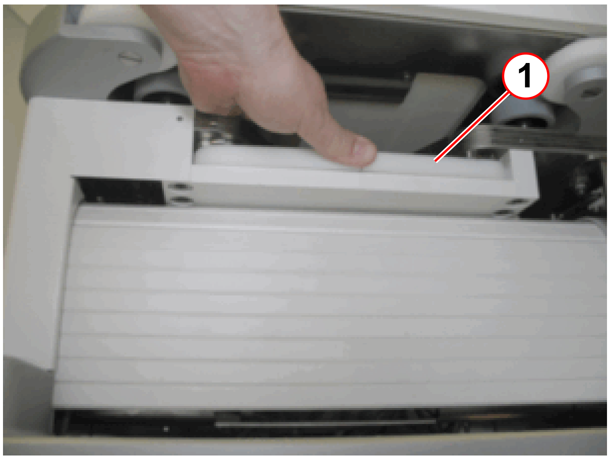
PM Option Combi Dockable Table: Checking the docking station

- Move the table approximately 300 mm into the magnet.
- Step on the docking / undocking and the up / down pedals.

  » Stepping on the pedals has no effect on the table.
- Move the table all the way out (home position).

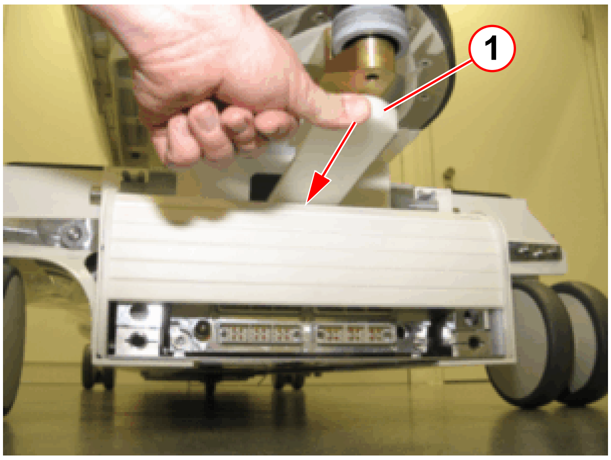
- Step on the docking / undocking and the up / down pedals.
  » The table is undocked.
- Slide back the roller blind (→ Fig. 33 Page 43)(→ Fig. 34 Page 43).
- Check the roller blind of the docking station and of the dockable table for wear and tear.
- Check the function and mobility of the roller blinds.

Fig. 33: Docking station - roller blind



(1) Lever to move the roller blind

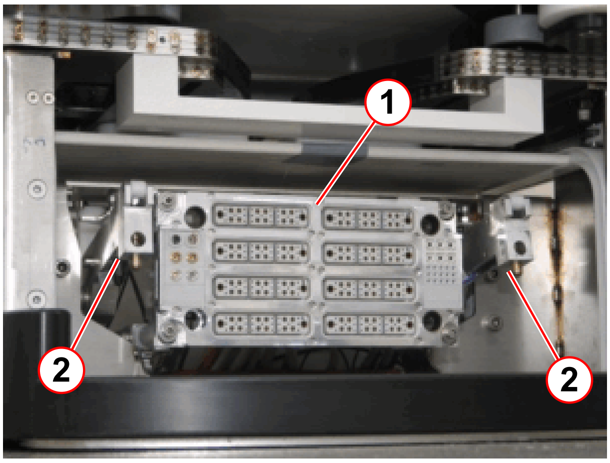
Fig. 34: Patient table - roller blind



(1) Lever to move the roller blind

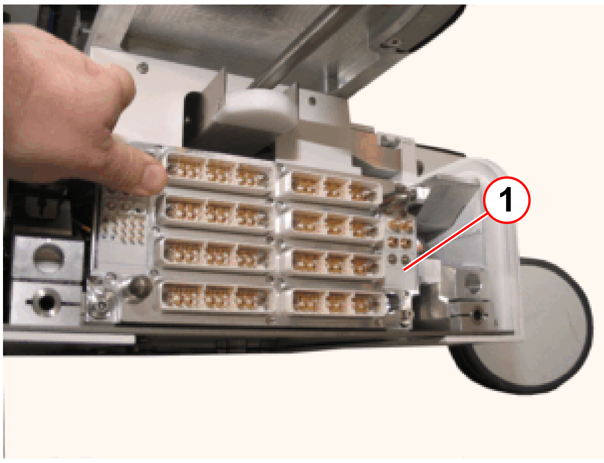
The connector assemblies are movable mounted

Fig. 35: Patient table docking port on the magnet



(1) Connector assembly
(2) Locking hook

Fig. 36: Patient table docking port



(1) Connector assembly

- Check the mobility of the connector assemblies (→ Fig. 35 Page 43).
  » The connector assemblies of the patient table and the docking station are free to move.
- Check the connectors for wear and tear.
  » The connectors must not be damaged

- Clean the connectors by using a brush and a non-fuzzing cloth.
- Check the locking hooks (➔ 2/Fig. 35 Page 43) for wear and tear.
  » The locking hooks must not be damaged.

2.14 End of Maintenance Instructions

The service covers which have been removed during safety checks have to be reinstalled before handing over to customer.

Initial Version

These instruction and the (→ Safety-Related Tests according to IEC 62353 / M7-010.833.05) replace the following document:

- Maintenance Instructions Combined Workflow

The Maintenance Instructions Combined Workflow was split in two documents:

- The maintenance instruction (these document) (→ M7-010.831.05 / Preventive Maintenance / Page 1)
- Safety- related test (→ Safety-Related Tests according to IEC 62353 / M7-010.833.05)

Chapter	Changes
All	The procedures in these instructions are taken over from the document 'Maintenance Instructions Combined Workflow' with minor changes. ■ Initial maintenance 18 months. ■ RF room door: Checking the door lock of the RF room, moved to Safety Related Test. ■ Phantoms: Checking the phantoms, moved to Safety Related Test. ■ Patient fan: Checking the air filter of the patient fan, moved to Safety Related Test.

There are no Hazard IDs in this document.

healthcare.siemens.com/services

Siemens Healthineers Headquarters
Siemens Healthcare GmbH
Henkestr. 127
91052 Erlangen
Germany
Telephone: +49 9131 84-0
siemens.com/healthineers

Print No.: M7-010.831.05.01.02 | Replaces: n.a.
Doc. Gen. Date: 12.15 | Language: English

siemens.com/healthineers